

Transformation of Gorkov's Equation for Type II Superconductors into Transport-Like Equations*

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The stationary behavior of type II superconductors is completely described by Gorkov's equations for a set of four Green's functions, supplemented by two self-consistency equations for gap parameter $\Delta(r)$ and vector potential $A(r)$. Expanding all quantities as usual at the Fermi surface and averaging over impurity positions, this set of equations is transformed into a simpler set for integrated Green's functions (which still contain much more information than is needed in most cases). The resulting equations, when linearized, yield essentially Lüders' transport equation for de Gennes' correlation function. The full equations contain all the known results and provide a promising starting point for numerical calculations.

The thermodynamic potential is constructed as a functional of the integrated Green's functions and the mean fields Δ and A and a variational principle is formulated which uses this functional. Finally, paramagnetic scatterers are included (in Born approximation) as an example for possible generalizations of the new equations.

Introduction

The fundament for almost all theoretical work on type II superconductors has been laid by GORKOV¹ with the formulation of a set of coupled equations for the normal and anomalous Green's functions. We shall not describe here the vast amount of theoretical results, which has been derived from these equations; several excellent review articles become now available². However, there are many blanks left in the chart of results; to name just one, the structure of a single vortex line at temperatures well below T_c is unknown. In principle these blanks could be filled in by solving Gorkov's equations and the selfconsistency equations simultaneously, but this task seems to be beyond the capacity of even the most modern computers.

In most cases, however, one does not need all the information contained in the Green's functions which solve these equations; for many purposes it would be sufficient to know only $\Delta(r)$ and $A(r)$. Part of the forbidding complexity of the equations is just due to this unnecessary

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¹ GORKOV, L. P.: Soviet Phys. JETP **9**, 1364 (1959); **10**, 998 (1960).

² See the articles by WERTHAMER, N. R., by A. L. FETTER and P. C. HOHENBERG and by K. MAKI in Superconductivity, edit. by R. PARKS. New York: M. Dekker (to be published).

information, and we show here, that it is possible to trade some of this information for a reduction of complexity: we derive equations, the solutions of which are the Green's functions, integrated over the energy variable. These new equations have a formal similarity to transport equations and are related to Lüders' transport equation for de Gennes' kernel of the linearized selfconsistency equation; in fact they are the generalization of the latter for the general nonlinear case.

In section 1 we consider Gorkov's equations for the Green's functions of the superconductor with general Fermi surface and impurities. In these equations the usual approximations, namely averaging over impurities and expanding all momentum dependent quantities to lowest relevant order in $k - k_F$ around the Fermi surface, are introduced.

In section 2 the resulting Gorkov's equations are transformed by mere matrix algebra into equations for new quantities which do not contain the energy variable. Then a formal solution of Gorkov's equations is given, by means of which the new quantities can be identified with the Green's functions integrated over the energy variable.

In section 3 the new equations are discussed. They permit an easy reproduction of many known results and are considered as a good starting point for numerical calculations in those regions of parameter values, which have not been accessible to analysis so far. It is then shown that a functional exists, the stationarity conditions of which yield the new equations and the selfconsistency conditions as the Euler-Lagrange equations. If the solutions of the new equations are inserted, this functional equals the thermodynamic potential of the superconductor and attains its minimum under variation of gap parameter and vector potential, when the selfconsistency conditions are satisfied.

In section 4 finally the possibilities of generalizations are discussed and the treatment of paramagnetic impurity scattering is given as an example.

1. Derivation of Gorkov's Equations for the Averaged Green's Functions

We shall use the notation and some of the methods of an earlier paper by the author³. We start with the following version of Gorkov's equations (Formula (1.5) of Ref. ³)^{*}:

$$\begin{pmatrix} i\omega + \mu - E\left(\mathbf{k} + \frac{\partial}{i}\right); & \Delta(\mathbf{r}) \\ -\Delta^*(\mathbf{r}); & -i\omega + \mu - E\left(\mathbf{k} + \frac{\partial}{i}\right) \end{pmatrix} \hat{\mathcal{G}}(\omega, \mathbf{k}, \mathbf{k}', \mathbf{r}) =$$

^{*} $\hbar = 1$ throughout.

³ EILENBERGER, G.: Z. Physik **190**, 142 (1966).

$$-\int \frac{d^3 q}{(2\pi)^3} V(\mathbf{k}-\mathbf{q}) \hat{\mathcal{G}}(\omega, \mathbf{q}, \mathbf{k}', \mathbf{r}) = \hat{1} (2\pi)^3 \delta^3(\mathbf{k}-\mathbf{k}'); \quad (1.1)$$

$$\hat{\mathcal{G}} = \begin{pmatrix} G & F \\ F^+ & G^+ \end{pmatrix}.$$

The function $E(\mathbf{k})$ is here the energy function of the electrons in a band; it is not necessary to restrict ourselves to $E(\mathbf{k}) = k^2/2m$ as was done in Ref. ³. The characteristic distances ξ, λ over which $A(\mathbf{r})$ and $A(\mathbf{r})$ vary appreciably are usually much larger than the electron wavelength $2\pi/k$ in the region of $\mathbf{k}$ -space close to the Fermi surface. We may put therefore (with Fermi energy μ)

$$E\left(\mathbf{k} + \frac{\partial}{i}\right) - \mu = \zeta(\mathbf{k}) + \mathbf{v}(\mathbf{k}) \frac{\partial}{i} \quad (1.2)$$

in this region. The energy width relevant for calculations is determined by the fact, that the Green's functions contribute appreciably to superconductivity only for $|\omega| \lesssim \omega_0$ (ω_0 is the characteristic phonon frequency) and that these Green's functions essentially vanish for $|\zeta| \gg \omega_0$.

For $|\zeta| \lesssim \omega_0$ it is usually a good approximation to replace $v(\mathbf{k})$ by $v(\mathbf{k}_F)$ in (1.2):

$$E\left(\mathbf{k} + \frac{\partial}{i}\right) - \mu = \zeta + \mathbf{v}(\mathbf{k}_F) \frac{\partial}{i} \quad (1.3)$$

where we characterize a vector $\mathbf{k}$ by $\zeta(\mathbf{k})$ and by the vector $\mathbf{k}_F$ of the point where the ray in direction $\mathbf{k}$ (in the reduced zone scheme) penetrates the Fermi surface $\zeta(\mathbf{k}) = 0$.

Another approximation,

$$\frac{d^3 k}{(2\pi)^3} = 2\pi N(0) d^2 k_F \rho(\mathbf{k}_F) \frac{d\zeta}{2\pi}, \quad (1.4)$$

is made, where

$$N(0) = \int_{S_F} \frac{d^2 k_F}{(2\pi)^3} \frac{1}{|\mathbf{v}(\mathbf{k}_F)|}$$

is the total normal density of states for one spin at the Fermi surface S_F and

$$\rho(\mathbf{k}_F) = ((2\pi)^3 N(0) |\mathbf{v}(\mathbf{k}_F)|)^{-1}$$

is the normal density of states for one spin at $\mathbf{k}_F$, normalized to

$$\int_{S_F} d^2 k_F \rho(\mathbf{k}_F) = 1.$$

This approximation, which is essential for the following, is usually quite excellent; it might fail, however, if the Fermi surface is dominated by

very fine structures, i.e., if $E(\mathbf{k})$ varies considerably over energy intervals ω_0 or momentum intervals $1/\xi$, $1/\lambda$.

The second basic point of our derivation is the standard averaging procedure over the impurity positions. It is known that this procedure yields a structure

$$\hat{\mathcal{G}}(\omega, \mathbf{k}, \mathbf{k}', \mathbf{r}) = (2\pi)^3 \delta^3(\mathbf{k} - \mathbf{k}') \hat{\mathcal{G}}(\omega, \mathbf{k}, \mathbf{r}) \quad (1.5)$$

for the averaged Green's functions.

In the case of a normal metal, one determines the averaged Green's function $G(\omega, \mathbf{k})$ from the integral equation

$$\overrightarrow{\overleftarrow{k}} = \overrightarrow{k} + \overrightarrow{k} \times \overleftarrow{k'} \times \overrightarrow{k} \quad (1.6)$$

where $\overrightarrow{\overleftarrow{k}}$ stands for $G(\omega, \mathbf{k})$ and $\overrightarrow{k}$ for the solution of the corresponding problem with $V=0$. (We always omit the term $\rightarrow \times \leftarrow$, which can be incorporated by a small shift in the Fermi energy.) For a superconductor, the same integral equation is valid³ (where the lines now stand for matrices $\hat{\mathcal{G}}$), as long as the term with ∂/i is absent in (1.1). If, however, one tries to expand the full equation (1.1) in powers of V , the operator ∂/i has to be applied to products of Green's functions thus leading to creation of additional terms.

For example, from

$$\hat{\mathcal{H}}_0 \hat{\mathcal{G}}_0(\omega, \mathbf{k}, \mathbf{r}) = \hat{1}$$

with

$$\hat{\mathcal{H}}_0 = \begin{pmatrix} i\omega - \zeta - \mathbf{v}(\mathbf{k}_F) \frac{\partial}{i}, & \Delta(\mathbf{r}) \\ -\Delta^*(\mathbf{r}); & -i\omega - \zeta - \mathbf{v}(\mathbf{k}_F) \frac{\partial}{i} \end{pmatrix},$$

follows

$$\begin{aligned} \hat{\mathcal{H}}_0 \left\{ \hat{\mathcal{G}}_0(\omega, \mathbf{k}, \mathbf{r}) \int \frac{d^3 q}{(2\pi)^3} V(\mathbf{k} - \mathbf{q}) \hat{\mathcal{G}}_0(\omega, \mathbf{q}, \mathbf{r}) \right\} \\ = \int \frac{d^3 q}{(2\pi)^3} V(\mathbf{k} - \mathbf{q}) \hat{\mathcal{G}}_0(\omega, \mathbf{q}, \mathbf{r}) \\ - \hat{\mathcal{G}}_0(\omega, \mathbf{k}, \mathbf{r}) \int \frac{d^3 q}{(2\pi)^3} V(\mathbf{k} - \mathbf{q}) \mathbf{v}(\mathbf{k}_F) \frac{\partial}{i} \hat{\mathcal{G}}_0(\omega, \mathbf{q}, \mathbf{r}). \end{aligned} \quad (1.7)$$

The appearance of terms like the last one gives rise to important and rather complicated "vertex corrections" in all types of calculations for impure superconductors^{2,3}. They can be avoided by considering operator Green's functions $\hat{\mathcal{G}}^{\text{op}}(\omega, \mathbf{k}, \mathbf{r})$ which are functions of $\omega, \mathbf{k}, \mathbf{r}$ and ∂/i and operate on functions in $\mathbf{r}$ space.

The operator $\hat{\mathcal{G}}_0^{\text{op}}(\omega, \mathbf{k}, \mathbf{r})$ is then defined by

$$\hat{\mathcal{H}}_0 \hat{\mathcal{G}}_0^{\text{op}}(\omega, \mathbf{k}, \mathbf{r}) = \hat{1}.$$

This means for arbitrary matrix functions $\hat{\mathcal{F}}(\mathbf{r})$

$$\hat{\mathcal{H}}_0 \hat{\mathcal{G}}_0^{\text{op}}(\omega, \mathbf{k}, \mathbf{r}) \hat{\mathcal{F}}(\mathbf{r}) = \hat{\mathcal{F}}(\mathbf{r}).$$

If one considers the equation corresponding to (1.7) but with $\hat{\mathcal{G}}_0$ now replaced by $\hat{\mathcal{G}}_0^{\text{op}}$, the last term at the r.h.s. no longer appears. Therefore the function $\hat{\mathcal{G}}^{\text{op}}(\omega, \mathbf{k}, \mathbf{r})$ appearing in the operator equation analogous to (1.5) obeys an integral equation of type (1.6), where the lines now stand for $\hat{\mathcal{G}}^{\text{op}}(\omega, \mathbf{k}, \mathbf{r})$ and $\hat{\mathcal{G}}_0^{\text{op}}(\omega, \mathbf{k}, \mathbf{r})$ respectively. Applying $\hat{\mathcal{H}}_0$ to this equation from the left finally yields

$$\left\{ \begin{pmatrix} i\omega - \zeta - \mathbf{v}(\mathbf{k}_F) \frac{\partial}{\partial i}; & \Delta(\mathbf{r}) \\ -\Delta^*(\mathbf{r}); & -i\omega - \zeta - \mathbf{v}(\mathbf{k}_F) \frac{\partial}{\partial i} \end{pmatrix} - \hat{\Sigma}^{\text{op}}(\omega, \mathbf{k}_F, \mathbf{r}) \right\} \hat{\mathcal{G}}^{\text{op}}(\omega, \zeta, \mathbf{k}_F, \mathbf{r}) = \hat{1} \quad (1.8)$$

where

$$\hat{\Sigma}^{\text{op}}(\omega, \mathbf{k}_F, \mathbf{r}) = \frac{1}{2} \int_{S_F} d^2 q_F \rho(\mathbf{q}_F) \mathcal{W}(\mathbf{k}_F, \mathbf{q}_F) \oint \frac{d\eta}{2\pi} \hat{\mathcal{G}}^{\text{op}}(\omega, \eta, \mathbf{q}_F, \mathbf{r}).$$

This is our final version of Gorkov's equations with all the standard approximations:

The function $\mathcal{W}(\mathbf{k}_F, \mathbf{q}_F)$ is the probability of scattering from a state $\mathbf{q}_F$ into a state $\mathbf{k}_F$ at the Fermi surface. Originally, the selfenergy in (1.8) read

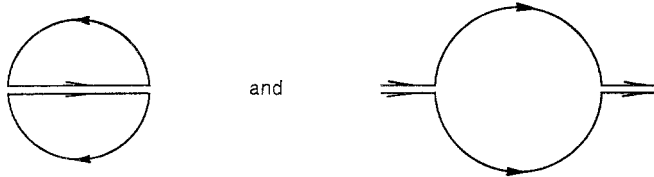
$$\int \frac{d^3 q}{(2\pi)^3} \mathcal{W}(\mathbf{k}, \mathbf{q}) \hat{\mathcal{G}}^{\text{op}}(\omega, \mathbf{q}, \mathbf{r}).$$

The assumption was made, then, that $\mathcal{W}(\mathbf{k}, \mathbf{q})$ varies sufficiently slowly in the relevant region of energies $\zeta(\mathbf{k}), \eta(\mathbf{q})$ to be replaced by $\mathcal{W}(\mathbf{k}_F, \mathbf{q}_F)$. However, the first term in the asymptotic expansion of $\hat{\mathcal{G}}^{\text{op}}$ is $\hat{1}\eta^{-1}$,

and therefore the integral $\int_{-\infty}^{+\infty} \hat{\mathcal{G}}^{\text{op}}(\eta) d\eta/2\pi$ does not converge. This problem is removed by the decomposition

$$\int_{-\infty}^{+\infty} \frac{d\eta}{2\pi} \dots = \frac{1}{2} \oint \frac{d\eta}{2\pi} \dots + \frac{1}{2} \int \frac{d\eta}{2\pi} \dots$$

where the contours of integration in the complex η -plane are



for $\oint$ and f respectively, and after integration the limit of large circles has to be taken.

In this limit the entire integral f comes from the term $1\eta^{-1}$ in the asymptotic expansion of $\hat{\mathcal{G}}$ and contributes to $\hat{\Sigma}$ a real, slowly varying function of $\mathbf{k}$ which is independent of ω , $\Delta(\mathbf{r})$ and $\mathbf{A}(\mathbf{r})$ and can be considered as a small renormalization of the function $E(\mathbf{k})$ due to the impurities. The integral $\oint$, on the other hand, is given by the residues of the poles of $\hat{\mathcal{G}}^{\text{op}}(\omega, \eta, \mathbf{q}_F, \mathbf{r})$ as function of η . For $|\omega| \lesssim \omega_0$ these are sufficiently close to $\eta=0$ to justify the replacement of $\mathcal{W}(\mathbf{k}, \mathbf{q})$ by $\mathcal{W}(\mathbf{k}_F, \mathbf{q}_F)$.

The function $\mathcal{W}(\mathbf{k}_F, \mathbf{q}_F)$, corresponding to the Born approximation implicit in (1.6), is given by

$$\mathcal{W}(\mathbf{k}_F, \mathbf{q}_F) = 2\pi N(0) n_i |u(\mathbf{k}_F - \mathbf{q}_F)|^2,$$

where n_i is the density of impurities and $u(\mathbf{k})$ the Fourier transformed impurity potential. For a spatially homogeneous superconductor it is relatively easy to show, that in the general case $u(\mathbf{k} - \mathbf{q})$ has to be replaced by the normal scattering t -matrix. This is certainly still a very good approximation for a superconductor with spatially varying $\Delta(\mathbf{r})$ although it is no longer the rigorous treatment of the non-Born-Scattering terms.

The selfconsistency equations, which constitute a closed system of equations together with (1.8) are

$$\frac{\Delta(\mathbf{r})}{\lambda} = \langle \psi_{\downarrow}(\mathbf{r}) \psi_{\uparrow}(\mathbf{r}) \rangle$$

(λ : Gorkov's coupling constant) and

$$\begin{aligned} \text{rot}(\mathbf{B}(\mathbf{r}) - \mathbf{B}_e(\mathbf{r})) &= \frac{4\pi}{c} \mathbf{j}_s(\mathbf{r}) \\ &= -4\pi \sum_{\alpha} \int d^3 r' \left\langle \frac{\delta}{\delta \mathbf{A}(\mathbf{r})} (\psi_{\alpha}^+(\mathbf{r}') E \psi_{\alpha}(\mathbf{r}')) \right\rangle \end{aligned}$$

with

$$E = E \left(\frac{\partial}{i \partial \mathbf{r}'} - \frac{e}{c} \mathbf{A}(\mathbf{r}') \right).$$

Expressing these by means of the elements of $\hat{\mathcal{G}}$ we get

$$\begin{aligned} &\Delta(\mathbf{r}) N(0) \ln \frac{T}{T_c} \\ &+ \frac{\pi N(0)}{\beta} \sum_{l=-\infty}^{+\infty} \left\{ \frac{\Delta(\mathbf{r})}{|\omega_l|} + \int_{S_F} d^2 k_F \rho(\mathbf{k}_F) \oint \frac{d\zeta}{2\pi} F(\omega_l, \zeta, \mathbf{k}_F, \mathbf{r}) \right\} = 0 \end{aligned} \quad (1.9)$$

and

$$\begin{aligned} & \frac{1}{4\pi} \operatorname{rot}(\mathbf{B}(\mathbf{r}) - \mathbf{B}_e(\mathbf{r})) \\ & - \frac{2e}{c} \frac{\pi N(0)}{\beta} \sum_{l=-\infty}^{+\infty} \int_{S_F} d^2 k_F \rho(\mathbf{k}_F) \mathbf{v}(\mathbf{k}_F) \oint \frac{d\zeta}{2\pi} G(\omega_l, \zeta, \mathbf{k}_F, \mathbf{r}) = 0 \end{aligned} \quad (1.10)$$

where

$$\omega_l = \frac{\pi}{\beta} (2l + 1).$$

λ is replaced by the standard cutoff procedure^{*}

$$\frac{1}{\lambda} \rightarrow N(0) \left(\ln \frac{T}{T_c} + \frac{\pi}{\beta} \sum_{l=-\infty}^{+\infty} \frac{1}{|\omega_l|} \right).$$

The energy integrals take the form $\oint$ because the term $\hat{1}\zeta^{-1}$ in the asymptotic expansion of $\hat{\mathcal{G}}$ contributes neither to (1.9) nor to (1.10).

2. Transformation into Simpler Equations

All integrals in (1.8)–(1.10) contain Green's functions only in the form

$$\oint \frac{d\zeta}{2\pi} \hat{\mathcal{G}}$$

(or some component of it). This is due to the fact that these expressions are derived from expectation values of fermion operators at the same position. Therefore, we can transform (1.8) into equations for

$$\oint \frac{d\zeta}{2\pi} \hat{\mathcal{G}},$$

the knowledge of which is sufficient for most purposes.

We introduce the vector^{**}

$$\vec{T} = (\hat{T}_1, \hat{T}_2, \hat{T}_3)$$

whose components are the Pauli matrices

$$\hat{T}_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{T}_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \hat{T}_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

^{*} Physically, the cutoff procedure consists in omitting all terms with $|\omega_l| > \omega_0$ in the l -sums; this was crucial in justifying certain approximations. Mathematically, however, the method used here is more convenient, it amounts to the same in the weak coupling case $\omega_0 \gg k_B T_c$, as now the terms with $|\omega_l| > \omega_0$ cancel each other. Then ω_0 as well as λ are contained in our equations only implicitly through T_c .

^{**} We distinguish the vectors $\mathbf{r}$, $\mathbf{k}$ etc., which are printed bold face from vectors of the type $\vec{a}$, $\vec{T}$ denoted by an arrow.

Then the operator $\hat{\mathcal{H}}$, to be applied to $\hat{\mathcal{G}}$ on the left hand side of (1.8), can be written as

$$\hat{\mathcal{H}} = - \left(\zeta + \mathbf{v}(\mathbf{k}_F) \frac{\partial}{i} \right) \hat{1} + i(\vec{w}_0 + \vec{w}_i) \vec{T} \quad (2.1)$$

where the vector $\vec{w}_0$ is given by

$$\vec{w}_0 = (\text{Im } \Delta(\mathbf{r}), \text{Re } \Delta(\mathbf{r}), \omega) \quad (2.2)$$

and $\vec{w}_i$ describes the impurity part of $\hat{\mathcal{H}}$:

$$\vec{w}_i(\omega, \mathbf{k}_F, \mathbf{r}) = \frac{1}{2} \int_{S_F} d^2 q_F \rho(\mathbf{q}_F) \mathcal{W}(\mathbf{k}_F, \mathbf{q}_F) \vec{a}_g(\omega, \mathbf{q}_F, \mathbf{r}). \quad (2.3)$$

Here we have used that

$$\oint \frac{d\eta}{2\pi} \hat{\mathcal{G}}^{\text{op}}(\omega, \eta, \mathbf{q}_F, \mathbf{r}) = -i \vec{a}_g(\omega, \mathbf{q}_F, \mathbf{r}) \vec{T} \quad (2.4)$$

which will be justified below.

Now let $\vec{a}$ be any vector with complex components and

$$\vec{a}^2 = 1. \quad (2.5)$$

Consider the matrices

$$(\hat{1} + \sigma \vec{a} \vec{T}), \quad \sigma = +1 \quad \text{or} \quad -1,$$

special matrices of this type have been introduced in Ref. ³. They have the interesting property to annihilate each other for $\sigma \neq \sigma'$:

$$(\hat{1} + \sigma \vec{a} \vec{T})(\hat{1} + \sigma' \vec{a} \vec{T}) = 2\delta_{\sigma\sigma'}(\hat{1} + \sigma \vec{a} \vec{T}). \quad (2.6)$$

Using these matrices we write $\hat{\mathcal{H}}$ in the form

$$\hat{\mathcal{H}} = \frac{1}{4} \sum_{\sigma} (\hat{1} + \sigma \vec{a} \vec{T}) \left(-\zeta - \mathbf{v}(\mathbf{k}_F) \frac{\partial}{i} + i \sigma W \right) (\hat{1} + \sigma \vec{a} \vec{T}) \quad (2.7)$$

where $\vec{a}$ and W are complex quantities yet to be determined; they can depend on the variables $\omega, \mathbf{k}_F, \mathbf{r}$ but not on ζ :

$$W = W(\omega, \mathbf{k}_F, \mathbf{r}); \quad \vec{a} = \vec{a}(\omega, \mathbf{k}_F, \mathbf{r}).$$

The idea behind trying the representation (2.7) is, that it suggests an easy formal solution of $\hat{\mathcal{H}} \hat{\mathcal{G}}^{\text{op}} = \hat{1}$ by using (2.6).

Comparing (2.7) with (2.1) term by term we get the condition

$$\vec{a} W - \frac{1}{2} \vec{a} \times \left(\mathbf{v}(\mathbf{k}_F) \frac{\partial}{i} \right) \vec{a} = \vec{w}_0 + \vec{w}_i. \quad (2.8)$$

In deriving this result we used the relation

$$\vec{a} \left(\mathbf{v}(\mathbf{k}_F) \frac{\partial}{i} \right) \vec{a} = \frac{1}{2} \left(\mathbf{v}(\mathbf{k}_F) \frac{\partial}{i} \right) \vec{a}^2 = 0 \quad \text{since } \vec{a}^2 = 1.$$

However, one has to be careful here since ∂/i denotes a “gauge invariant” differentiation³. Applied to matrices of the type $\vec{a} \vec{T}$, it is defined as

$$\begin{aligned} \frac{\partial}{i} \vec{a} \vec{T} &= \hat{T}_3 \frac{\partial}{i \partial \mathbf{r}} a_3 \\ &+ \frac{1}{2} \left\{ (\hat{T}_1 + i \hat{T}_2) \left(\frac{\partial}{i \partial \mathbf{r}} - \frac{2e}{c} \mathbf{A}(\mathbf{r}) \right) (a_1 - i a_2) \right. \\ &\quad \left. + (\hat{T}_1 - i \hat{T}_2) \left(\frac{\partial}{i \partial \mathbf{r}} + \frac{2e}{c} \mathbf{A}(\mathbf{r}) \right) (a_1 + i a_2) \right\} \\ &= \left(\frac{\partial}{i \partial \mathbf{r}} \vec{a} - i \frac{2e}{c} \mathbf{A}(\mathbf{r}) \vec{e}_3 \times \vec{a} \right) \vec{T} \end{aligned} \quad (2.9)$$

(with unit vector $\vec{e}_3$ in 3-direction).

Taking the scalar and the vector product of (2.8) with $\vec{a}$ and using (2.9) again we arrive at

$$W(\omega, \mathbf{k}_F, \mathbf{r}) = \vec{a}(\omega, \mathbf{k}_F, \mathbf{r}) (\vec{w}_0(\omega, \mathbf{r}) + \vec{w}_i(\omega, \mathbf{k}_F, \mathbf{r})) \quad (2.10)$$

and

$$\left(\mathbf{v}(\mathbf{k}_F) \frac{\partial}{i} \right) \vec{a}(\omega, \mathbf{k}_F, \mathbf{r}) = 2 \vec{a}(\omega, \mathbf{k}_F, \mathbf{r}) \times (\vec{w}_0(\omega, \mathbf{r}) + \vec{w}_i(\omega, \mathbf{k}_F, \mathbf{r})). \quad (2.11)$$

This is essentially the equation we wanted to derive. We postpone its discussion to the next section and investigate $\hat{\mathcal{G}}$ first.

We try to solve $\hat{\mathcal{H}} \hat{\mathcal{G}}^{\text{op}} = \hat{1}$ in the form

$$\hat{\mathcal{G}}^{\text{op}} = \frac{1}{4} \sum_{\sigma} (\hat{1} + \sigma \vec{a} \vec{T}) \frac{1}{-\hat{1} \left(\zeta + \mathbf{v}(\mathbf{k}_F) \frac{\partial}{i} - i \sigma W \right) + \hat{R}_{\sigma}} (\hat{1} + \sigma \vec{a} \vec{T}) \quad (2.12)$$

where $\hat{R}_{\sigma} = \hat{R}(\sigma, \omega, \mathbf{k}_F, \mathbf{r})$ is diagonal, i.e., contains $\hat{1}$ and $\hat{T}_3$ only. Then, using the all-important relation (2.6) we get

$$\hat{\mathcal{H}} \hat{\mathcal{G}}^{\text{op}} = \hat{1} - \frac{1}{4} \sum_{\sigma} (\hat{1} + \sigma \vec{a} \vec{T}) \left(\frac{\sigma}{2} \left(\mathbf{v} \frac{\partial}{i} \right) \vec{a} \vec{T} + \hat{R}_{\sigma} \right) \frac{1}{\hat{D}} (\hat{1} + \sigma \vec{a} \vec{T}) \quad (2.13)$$

where $\hat{D}$ abbreviates the denominator of (2.12).

To get rid of the second term in (2.13) we require

$$(\hat{1} + \sigma \vec{a} \vec{T}) \left(\frac{\sigma}{2} \left(\mathbf{v} \frac{\partial}{\partial} \right) \vec{a} \vec{T} + \hat{R}_\sigma \right) = 0 \quad (2.14)$$

for both values of σ . Indeed; after some elementary algebra one arrives at

$$\hat{R}_\sigma = (a_3 \hat{1} - \sigma \hat{T}_3) \frac{\left(\mathbf{v} \frac{\partial}{\partial} \right) a_3 + i \sigma \vec{e}_3 \left(\vec{a} \times \left(\mathbf{v} \frac{\partial}{\partial} \right) \vec{a} \right)}{2(a_1^2 + a_2^2)}. \quad (2.15)$$

We believe that usually $\hat{R}_\sigma$ is small compared to W , which in the limit of constant Δ equals the quantity

$$\sqrt{\omega^2 + |\Delta|^2} + \frac{1}{2\tau},$$

familiar from calculations for type II superconductors. It is interesting to notice, however, that (2.15) contains contributions of order zero in $\Delta(\mathbf{r})$ if one solves (2.11) by a series expansion in powers of Δ . This indicates that $\hat{\mathcal{G}}^{\text{op}}$ cannot be expanded rigorously in a power series in Δ even if $\vec{a}$ can be.

Next we have to evaluate the integral

$$\oint \frac{d\zeta}{2\pi} \hat{\mathcal{G}}^{\text{op}}(\omega, \zeta, \mathbf{k}_F, \mathbf{r}) = -\frac{1}{4} \sum_\sigma (\hat{1} + \sigma \vec{a} \vec{T}) \left\{ \oint \frac{d\zeta}{2\pi} \frac{1}{\left(\zeta + \mathbf{v} \frac{\partial}{\partial} - i \sigma W \right) \hat{1} - \hat{R}_\sigma} \right\} (\hat{1} + \sigma \vec{a} \vec{T}). \quad (2.16)$$

We make the assumption, discussed below, that the zeros of the denominator as function of ζ lie in the half plane $\sigma \text{Im } \zeta > 0$. Then the integral in (2.16) yields immediately $i\sigma \hat{1}$ and we get

$$\oint \frac{d\zeta}{2\pi} \hat{\mathcal{G}}^{\text{op}}(\omega, \zeta, \mathbf{k}_F, \mathbf{r}) = -i \vec{a}(\omega, \mathbf{k}_F, \mathbf{r}) \vec{T} = - \begin{pmatrix} i a_3 & a_2 + i a_1 \\ -a_2 + i a_1 & -i a_3 \end{pmatrix} \quad (2.17)$$

thus justifying (2.4) with $\vec{a}_g = \vec{a}$.

With (2.3), (2.11) and (2.17) we get the final equation for $\vec{a}$:

$$\left(\mathbf{v}(\mathbf{k}_F) \frac{\partial}{\partial} \right) \vec{a}(\omega, \mathbf{k}_F, \mathbf{r}) = \vec{a}(\omega, \mathbf{k}_F, \mathbf{r}) \times \left\{ 2 \vec{w}_0(\omega, \mathbf{r}) + \int_{S_F} d^2 q_F \rho(\mathbf{q}_F) \mathcal{W}(\mathbf{k}_F, \mathbf{q}_F) \vec{a}(\omega, \mathbf{q}_F, \mathbf{r}) \right\}. \quad (2.18)$$

The poles of the integrand in (2.16) would be given by the complex eigenvalues of the operators in the diagonal elements of

$$\begin{aligned} & \left(v(k_F) \frac{\partial}{i} - i \sigma W(\omega, k_F, r) \right) \hat{1} - \hat{R}_\sigma(\omega, k_F, r) \\ &= \left(v(k_F) \frac{\partial}{i} \right) \hat{1} - \hat{W}_R(\sigma, \omega, k_F, r) - i \hat{W}_I(\sigma, \omega, k_F, r) \end{aligned}$$

(with real diagonal matrix functions $\hat{W}_R, \hat{W}_I$), if these operators would only have a complete orthogonal set of eigenfunctions. This is, however, not the case since the imaginary part, $i \hat{W}_I(\sigma, \omega, k_F, r)$, in general varies with position and therefore the operator does not commute with its adjoint. This is why we have not succeeded in proving rigorously the assumption leading to (2.17); it is, however, very plausible as can be seen by the following arguments:

If, for instance, the integrand is expanded in a power series in $v \partial / i$:

$$-\frac{1}{\hat{D}} = \frac{1}{(\zeta - i \sigma W) \hat{1} - \hat{R}_\sigma} - \frac{1}{(\zeta - i \sigma W) \hat{1} - \hat{R}_\sigma} v \frac{\partial}{i} \frac{1}{(\zeta - i \sigma W) \hat{1} - \hat{R}_\sigma} + \dots$$

and if this series converges uniformly on the entire path of ζ -integration, then only the first term of this expansion contributes to the integral and gives i or $-i$ depending upon the signs of the matrix elements of $\hat{W}_I$.

The same result holds, if $\hat{W}_I(\sigma, \omega, k_F, r)$ does not deviate too much from an expression $\hat{W}_{I_0}(\sigma, \omega, k_F)$ independent of r . Then one can expand $\hat{D}^{-1}$ in powers of $\hat{W}_I - \hat{W}_{I_0}$, getting now denominators

$$\left(\zeta + v \frac{\partial}{i} \right) \hat{1} - \hat{W}_R - i \hat{W}_{I_0},$$

which are normal operators with eigenvalues $\zeta \hat{1} + i \hat{W}_{I_0} + \text{something real}$.

It is therefore reasonable to assume that the position of the poles is given by the sign of the imaginary part of $i \sigma W \hat{1} + \hat{R}_\sigma$. We assume further that this imaginary part is dominated by the term proportional to $i \sigma \hat{1}$ in the denominator:

$$W + \frac{a_3 \vec{e}_3 \left(\vec{a} \times \left(v \frac{\partial}{i} \right) \vec{a} \right)}{2(a_1^2 + a_2^2)} \quad (2.19)$$

and that the real part of (2.19) does not change sign along the lines $r(y)=r_0+vy$ for any given value of the parameters ω , k_F and r_0 . If one solves the differential Eq. (2.11) along such a line, $\tilde{a}$ is determined only up to a sign and we shall now fix this sign so as to make the real part of (2.19) positive, thus getting the desired position of the poles. The validity of the last two assumptions should be checked, of course, for every solution of (2.18).

3. Discussion of the New Equations

The notations adopted so far have proved to be very convenient for the derivation of (2.18). For a discussion and for practical purposes, however, a different form seems to be preferable. We define

$$\begin{aligned} g(\omega, k_F, r) &= a_3(\omega, k_F, r) = i \oint \frac{d\zeta}{2\pi} G(\omega, \zeta, k_F, r) \\ f(\omega, k_F, r) &= a_2(\omega, k_F, r) + i a_1(\omega, k_F, r) = -\oint \frac{d\zeta}{2\pi} F(\omega, \zeta, k_F, r) \quad (3.1) \\ f^+(\omega, k_F, r) &= a_2(\omega, k_F, r) - i a_1(\omega, k_F, r) = \oint \frac{d\zeta}{2\pi} F^+(\omega, \zeta, k_F, r). \end{aligned}$$

Using (2.9) we get from (2.18)

$$\begin{aligned} & \left\{ 2\omega + v(k_F) \left(\frac{\partial}{\partial r} - i \frac{2e}{c} A(r) \right) \right\} f(\omega, k_F, r) \\ &= 2A(r) g(\omega, k_F, r) + \int_{S_F} d^2 q_F \rho(q_F) \mathcal{W}(k_F, q_F) \\ & \quad \cdot \{ g(\omega, k_F, r) f(\omega, q_F, r) - f(\omega, k_F, r) g(\omega, q_F, r) \}, \end{aligned} \quad (3.2)$$

$$\begin{aligned} & \left\{ 2\omega - v(k_F) \left(\frac{\partial}{\partial r} + i \frac{2e}{c} A(r) \right) \right\} f^+(\omega, k_F, r) \\ &= 2A^*(r) g(\omega, k_F, r) + \int_{S_F} d^2 q_F \rho(q_F) \mathcal{W}(k_F, q_F) \\ & \quad \cdot \{ g(\omega, k_F, r) f^+(\omega, q_F, r) - f^+(\omega, k_F, r) g(\omega, q_F, r) \}. \end{aligned} \quad (3.3)$$

From $\tilde{a}^2=1$ follows

$$g(\omega, k_F, r) = (1 - f(\omega, k_F, r) f^+(\omega, k_F, r))^{\frac{1}{2}} \quad (3.4)$$

where the sign of the root is determined by the convention described at the end of section 2.

The system of equations is completed by the selfconsistency conditions

$$\Delta(r) N(0) \ln \frac{T}{T_c} + \frac{2\pi N(0)}{\beta} \sum_{l=0}^{\infty} \left\{ \frac{\Delta(r)}{\omega_l} - \int_{S_F} d^2 k_F \rho(\mathbf{k}_F) f(\omega_l, \mathbf{k}_F, \mathbf{r}) \right\} = 0, \quad (3.5)$$

$$\frac{1}{4\pi} \text{rot}(\mathbf{B}(\mathbf{r}) - \mathbf{B}_e(\mathbf{r})) + \frac{2ie}{c} \frac{2\pi N(0)}{\beta} \sum_{l=0}^{\infty} \int_{S_F} d^2 k_F \rho(\mathbf{k}_F) \mathbf{v}(\mathbf{k}_F) g(\omega_l, \mathbf{k}_F, \mathbf{r}) = 0. \quad (3.6)$$

In deriving the last two equations from (1.9) and (1.10) we have used the symmetry relations, which follow from (3.2)–(3.5):

$$\begin{aligned} f^*(-\omega, \mathbf{k}_F, \mathbf{r}) &= f^+(\omega, \mathbf{k}_F, \mathbf{r}); & g^*(-\omega, \mathbf{k}_F, \mathbf{r}) &= -g(\omega, \mathbf{k}_F, \mathbf{r}) \\ f^*(\omega, -\mathbf{k}_F, \mathbf{r}) &= f^+(\omega, \mathbf{k}_F, \mathbf{r}); & g^*(\omega, -\mathbf{k}_F, \mathbf{r}) &= g(\omega, \mathbf{k}_F, \mathbf{r}) \\ f(-\omega, -\mathbf{k}_F, \mathbf{r}) &= f(\omega, \mathbf{k}_F, \mathbf{r}); & g(-\omega, -\mathbf{k}_F, \mathbf{r}) &= -g(\omega, \mathbf{k}_F, \mathbf{r}) \end{aligned} \quad (3.7)$$

and thus restricted the l -summation in (3.5) and (3.6) to positive values of l .

The differential equation for $g(\omega, \mathbf{k}_F, \mathbf{r})$ which follows from (2.18),

$$\begin{aligned} 2\mathbf{v}(\mathbf{k}_F) \frac{\partial}{\partial \mathbf{r}} g(\omega, \mathbf{k}_F, \mathbf{r}) \\ = 2\Delta^*(\mathbf{r}) f(\omega, \mathbf{k}_F, \mathbf{r}) - 2\Delta(\mathbf{r}) f^+(\omega, \mathbf{k}_F, \mathbf{r}) + \int_{S_F} d^2 q_F \rho(\mathbf{q}_F) \\ \cdot \mathcal{W}(\mathbf{k}_F, \mathbf{q}_F) \{ f(\omega, \mathbf{k}_F, \mathbf{r}) f^+(\omega, \mathbf{q}_F, \mathbf{r}) - f^+(\omega, \mathbf{k}_F, \mathbf{r}) f(\omega, \mathbf{q}_F, \mathbf{r}) \} \end{aligned} \quad (3.8)$$

is redundant; it is also a consequence of (3.2)–(3.4) as it must be, and needs not be considered separately.

The Eqs. (3.2)–(3.4) are equivalent to Gorkov's equation (1.8); they are simpler, because they do not contain the variable ζ and have only two unknowns instead of four. Moreover, they suggest an intuitive physical interpretation by virtue of their formal similiarity with Laplace

transformed transport equations: 2ω can be interpreted as the transformed d/dt , $v(k_F) \partial/\partial r$ resembles the convective term, and, on the right hand side, there appear the in and out scattering terms as well as an inhomogeneity describing sources and sinks. In addition to the strange nonlinearity, the difference between these and genuine transport equations lies in the fact, that f and f^+ cannot be interpreted as probabilities; a complex $A(r)$ as well as the presence of a vector potential make these functions essentially complex so that they are capable to interfere.

If one linearizes the equation by putting $g=1$ (for positive ω), one gets

$$\left\{ 2\omega + \frac{1}{\tau} + v(k_F) \left(\frac{\partial}{\partial r} - i \frac{2e}{c} A(r) \right) \right\} f(\omega, k_F, r) \\ = 2A(r) + \int_{S_F} d^2 q_F \rho(q_F) \mathcal{W}(k_F, q_F) f(\omega, q_F, r)$$

with

$$\frac{1}{\tau} = \int_{S_F} d^2 q_F \rho(q_F) \mathcal{W}(k_F, q_F).$$

The Green's function (in the mathematical sense of this term) belonging to this inhomogeneous linear equation is closely related to de Gennes' correlation function⁴ appearing in the linearized gap equation which has been studied intensively by LÜDERS⁵ and collaborators.

From (3.2)–(3.6) many known results can be rederived in a relatively easy and transparent manner. The BCS result, Ginzburg-Landau and generalized Ginzburg-Landau equations are easily shown to follow. Especially the so called vertex corrections, which appear in approximate solutions of Gorkov's equations and usually require rather tedious considerations are automatically included in the nonlinear scattering term.

The new equations are, however, not only of intrinsic theoretical interest; but they also provide a basis for numerical calculations, more convenient than the original Gorkov's equations; some work in this direction has been started.

There remains open, however, the question of boundary conditions belonging to our differential equations. One would like to require that all functions remain bounded at infinity and that at finite boundaries some

⁴ GENNES, P. G. DE: Rev. Mod. Phys. **36**, 225 (1964); — Phys. Kond. Materie **3**, 79 (1967). — GENNES, P. G. DE, and M. TINKHAM: Physics **1**, 107 (1964).

⁵ LÜDERS, G.: Z. Naturforsch. **21a**, 680, 1415, 1425, 1842 (1966); **22a**, 865 (1967).

phenomenological conditions could be used, based on the transport interpretation of the equations as has been done in the linearized case⁶. We have not investigated the mathematical compatibility of such conditions as it seems to be difficult to do this on general grounds. It has to be done separately for each special application of the equations.

To derive an expression of the free energy in terms of the functions f and f^+ , we consider the quantity

$$\Omega = \int d^3 r \left\{ \frac{(\mathbf{B}(\mathbf{r}) - \mathbf{B}_e(\mathbf{r}))^2}{8\pi} + |\Delta(\mathbf{r})|^2 N(0) \ln \frac{T}{T_c} + \frac{2\pi N(0)}{\beta} \sum_{l=0}^{\infty} \left(\frac{|\Delta(\mathbf{r})|^2}{\omega_l} - \int_{S_F} d^2 k_F \rho(\mathbf{k}_F) I(\omega_l, \mathbf{k}_F, \mathbf{r}) \right) \right\} \quad (3.9)$$

where* (omitting arguments $\omega, \mathbf{k}_F, \mathbf{r}$ for brevity, where no confusion can arise):

$$I(\omega, \mathbf{k}_F, \mathbf{r}) = \Delta^* f + \Delta f^+ + 2\omega(g-1) + g v \left(\frac{1}{2} \frac{\partial}{\partial \mathbf{r}} \ln \frac{f}{f^+} - i \frac{2e}{c} \mathbf{A} \right) + \frac{1}{2} \int_{S_F} d^2 q_F \rho(\mathbf{q}_F) \mathcal{W}(\mathbf{k}_F, \mathbf{q}_F) \cdot \left\{ \frac{1}{2} f(\mathbf{k}_F) f^+(\mathbf{q}_F) + \frac{1}{2} f^+(\mathbf{k}_F) f(\mathbf{q}_F) + g(\mathbf{k}_F) g(\mathbf{q}_F) - 1 \right\}. \quad (3.10)$$

If we treat $f^+, f, \Delta^*, \mathbf{A}$ as arbitrary functions (g is now an abbreviation for $(1 - f f^+)^{\frac{1}{2}}$) and require stationarity of Ω under variation of these functions, we obtain Eqs. (3.2), (3.3), (3.5) and (3.6) respectively as Euler-Lagrange equations. For approximate calculations it would be very useful if under free variation of f^+, f, Δ and $\mathbf{A}$ the functional would have its absolute minimum or maximum for the correct solutions of (3.2) to (3.6).

Unfortunately, this is not true; but a weaker statement holds: for any given functions $\Delta(\mathbf{r})$ and $\mathbf{A}(\mathbf{r})$ we consider the solutions f and f^+ of (3.2) and (3.3) and insert these into (3.9), (3.10), thus obtaining a functional $\tilde{\Omega}(\Delta, \mathbf{A})$ of $\Delta(\mathbf{r})$ and $\mathbf{A}(\mathbf{r})$ alone. This functional as we show below, is identical with the difference $\Omega_s - \Omega_n$ between the thermodynamic potentials of the superconducting and normal phase and consequently has its absolute minimum for those functions $\Delta(\mathbf{r})$ and $\mathbf{A}(\mathbf{r})$ which satisfy (3.5) and (3.6).

* It is amusing to notice, that $\frac{1}{2} I(\omega, \mathbf{k}_F, \mathbf{r}) + \omega + (1/4\tau)$ equals (2.19), if in the latter the scattering term is multiplied by $\frac{1}{2}$.

⁶ LÜDERS, G.: Z. Physik **202**, 8 (1967). — USADEL, K.: To be published.

This difference $\Omega_s - \Omega_n$ for given $\Delta(\mathbf{r})$ and $\mathbf{A}(\mathbf{r})$ can be written as (see for instance⁷)

$$\Omega_s - \Omega_n = \int d^3r \left\{ \frac{|\Delta(\mathbf{r})|^2}{\lambda} + \frac{(\mathbf{B}(\mathbf{r}) - \mathbf{B}_e(\mathbf{r}))^2}{8\pi} \right\} - \frac{1}{\beta} \{ \ln \text{Tr} e^{-\beta \mathcal{H}_s(\Delta, \mathbf{A})} - \ln \text{Tr} e^{-\beta \mathcal{H}_n(\Delta, \mathbf{A})} \} \quad (3.11)$$

with the effective Hamiltonian

$$\mathcal{H}_s(\Delta, \mathbf{A}) = \mathcal{H}_n(\mathbf{A}) - \int d^3r \{ \Delta(\mathbf{r}) \psi_{\uparrow}^+(\mathbf{r}) \psi_{\downarrow}^+(\mathbf{r}) + \Delta^*(\mathbf{r}) \psi_{\downarrow}(\mathbf{r}) \psi_{\uparrow}(\mathbf{r}) \}$$

where $\mathcal{H}_n(\mathbf{A})$ is the one particle Hamiltonian of the normal electrons in the presense of the vector potential $\mathbf{A}(\mathbf{r})$. Now, if $f = f(\Delta, \mathbf{A})$ and $f^+ = f^+(\Delta, \mathbf{A})$ obey the Eqs. (3.2) and (3.3) for given functions $\Delta(\mathbf{r})$ and $\mathbf{A}(\mathbf{r})$, they represent the Green's functions belonging to $\mathcal{H}_s(\Delta, \mathbf{A})$; the corresponding expectation values $\langle \psi_{\downarrow}(\mathbf{r}) \psi_{\uparrow}(\mathbf{r}) \rangle$ and $\langle \mathcal{H}_s(\mathbf{r}) \rangle$ are then expressed in the usual manner by f and f^+ . For these f, f^+ the equations

$$\left. \frac{\delta}{\delta \Delta^*(\mathbf{r})} (\Omega_s - \Omega_n) \right|_{\mathbf{A}} = \text{l.h.s. of (3.5)} = \left. \frac{\delta}{\delta \Delta^*(\mathbf{r})} \Omega \right|_{\mathbf{A}, f, f^+} = \left. \frac{\delta}{\delta \Delta^*(\mathbf{r})} \tilde{\Omega} \right|_{\mathbf{A}} \quad (3.12)$$

$$\left. \frac{\delta}{\delta \mathbf{A}(\mathbf{r})} (\Omega_s - \Omega_n) \right|_{\Delta} = \text{l.h.s. of (3.6)} = \left. \frac{\delta}{\delta \mathbf{A}(\mathbf{r})} \Omega \right|_{\Delta, f, f^+} = \left. \frac{\delta}{\delta \mathbf{A}(\mathbf{r})} \tilde{\Omega} \right|_{\Delta} \quad (3.13)$$

hold true; where the last identities in (3.12) and (3.13) are valid because (3.2) and (3.3) are equivalent to

$$\left. \frac{\delta}{\delta f^+} \Omega \right|_{\Delta, \Delta, f} = 0 \quad \text{and} \quad \left. \frac{\delta}{\delta f} \Omega \right|_{\Delta, \Delta, f^+} = 0$$

respectively.

The variation of $f(\Delta, \mathbf{A})$ and $f^+(\Delta, \mathbf{A})$ therefore does not contribute to the variation of

$$\tilde{\Omega}(\Delta, \mathbf{A}) = \Omega(\Delta, \mathbf{A}, f(\Delta, \mathbf{A}), f^+(\Delta, \mathbf{A})).$$

Finally both $\tilde{\Omega}$ and $\Omega_s - \Omega_n$ vanish when Δ and $\mathbf{B}(\mathbf{r}) - \mathbf{B}_e(\mathbf{r})$ vanish, and, this together with (3.12) and (3.13) proves the identity of $\tilde{\Omega}$ and $\Omega_s - \Omega_n$ as functionals of $\Delta(\mathbf{r})$ and $\mathbf{A}(\mathbf{r})$.

⁷ EILENBERGER, G.: Z. Physik **184**, 427 (1965).

4. Inclusion of Paramagnetic Scatterers

Finally one may wish to incorporate in our equations spin paramagnetism, orbital diamagnetism, spin orbit scattering, or other effects which have been left out so far but become important under special circumstances. We treat in the following the case of paramagnetic scatterers. It ought to be possible to proceed similarly in other cases.

For the usual treatment of paramagnetic scattering, when one does not want to enter into the difficulties of the Kondo problem, one can start by adding a scattering term

$$\int d^3 r \sum_j u_p(\mathbf{r} - \mathbf{r}_j) \sum_{\alpha\beta} \vec{e}_j \vec{T}_{\alpha\beta} \psi_\alpha^+(\mathbf{r}) \psi_\beta(\mathbf{r}) \quad (4.1)$$

to the Hamiltonian, where the positions $\mathbf{r}_j$ and the directions of the unit vectors $\vec{e}_j$ are distributed at random. Since (4.1) does not commute with the total electron spin one has to introduce a larger Green's function matrix

$$\hat{\mathcal{G}} = \begin{pmatrix} G_{\downarrow\downarrow} & F_{\downarrow\uparrow} & G_{\uparrow\uparrow} & F_{\downarrow\downarrow} \\ F_{\uparrow\downarrow}^+ & G_{\uparrow\uparrow}^+ & F_{\uparrow\uparrow}^+ & G_{\uparrow\downarrow}^+ \\ G_{\uparrow\downarrow} & F_{\uparrow\uparrow} & G_{\uparrow\uparrow} & F_{\uparrow\downarrow} \\ F_{\downarrow\downarrow}^+ & G_{\downarrow\uparrow}^+ & F_{\downarrow\uparrow}^+ & G_{\downarrow\downarrow}^+ \end{pmatrix}. \quad (4.2)$$

(Similar generalizations would occur in the other cases mentioned above). The steps leading to (1.8) can now be repeated for the matrix $\hat{\mathcal{G}}$, yielding the equation*:

$$\left\{ \begin{pmatrix} -\hat{1} \left(\zeta + \mathbf{v}(\mathbf{k}_F) \frac{\partial}{\partial i} \right) + i \vec{w}_0 \vec{T}; & \hat{0} \\ \hat{0}; & -\hat{1} \left(\zeta + \mathbf{v}(\mathbf{k}_F) \frac{\partial}{\partial i} \right) + i \vec{w}_0 \vec{T} \end{pmatrix} - \hat{\Sigma}^{\text{op}}(\omega, \mathbf{k}_F, \mathbf{r}) \right\} \cdot \hat{\mathcal{G}}^{\text{op}}(\omega, \zeta, \mathbf{k}_F, \mathbf{r}) = \hat{1} \quad (4.3)$$

where

$$\begin{aligned} \hat{\Sigma}^{\text{op}}(\omega, \mathbf{k}_F, \mathbf{r}) = & \frac{1}{2} \int_{S_F} d^2 q_F \rho(\mathbf{q}_F) \oint \frac{d\eta}{2\pi} \left\{ \mathcal{W}(\mathbf{k}_F, \mathbf{q}_F) \hat{\mathcal{G}}^{\text{op}}(\omega, \eta, \mathbf{q}_F, \mathbf{r}) \right. \\ & + \mathcal{W}_p(\mathbf{k}_F, \mathbf{q}_F) \frac{1}{N_p} \sum_j \begin{pmatrix} e_{j3} \hat{T}_3; & (e_{j1} - i e_{j2}) \hat{1} \\ (e_{j1} + i e_{j2}) \hat{1}; & -e_{j3} \hat{T}_3 \end{pmatrix} \\ & \left. \cdot \hat{\mathcal{G}}^{\text{op}}(\omega, \eta, \mathbf{q}_F, \mathbf{r}) \begin{pmatrix} e_{j3} \hat{T}_3; & (e_{j1} - i e_{j2}) \hat{1} \\ (e_{j1} + i e_{j2}) \hat{1}; & -e_{j3} \hat{T}_3 \end{pmatrix} \right\}. \end{aligned} \quad (4.4)$$

* The vector $\vec{w}$ is defined to have the components $-w_1, -w_2, w_3$ if $\vec{w}$ has the components w_1, w_2, w_3 .

Here $\mathcal{W}_p(\mathbf{k}_F, \mathbf{q}_F)$ is the paramagnetic scattering probability in Born approximation:

$$\mathcal{W}_p(\mathbf{k}_F, \mathbf{q}_F) = 2\pi N(0) n_p |u_p(\mathbf{k}_F - \mathbf{q}_F)|^2. \quad (4.5)$$

In order to repeat the calculations of section 2 for a general 4×4 matrix equation one would have to use a general set of four matrices $\hat{M}_\sigma$, $\sigma = 1, 2, 3, 4$ which annihilate each other. This would, however, greatly increase the number of unknowns and equations corresponding to $\vec{a}(\omega, \mathbf{k}_F, \mathbf{r})$ and Eq. (2.18) respectively. It can be avoided in our case, because

$$\begin{pmatrix} \hat{0} & \hat{T}_3 \\ \hat{T}_3 & \hat{0} \end{pmatrix} \hat{\mathcal{G}} \begin{pmatrix} \hat{0} & \hat{T}_3 \\ \hat{T}_3 & \hat{0} \end{pmatrix}$$

obeys the same equation as $\hat{\mathcal{G}}$. This is easily shown by multiplying (4.3) with $\begin{pmatrix} \hat{0} & \hat{T}_3 \\ \hat{T}_3 & \hat{0} \end{pmatrix}$ from both sides and using the fact that the directions of $\vec{e}_j$ are randomly distributed*. It follows then, that the matrices $\hat{1} + \sigma \vec{a} \vec{T}$ of section 2 are to be replaced by

$$\begin{pmatrix} \hat{1} + \sigma \vec{a} \vec{T} & \hat{0} \\ \hat{0} & \hat{1} + \sigma \vec{a} \vec{T} \end{pmatrix}$$

for the present consideration. We get finally

$$\oint \frac{d\zeta}{2\pi} \hat{\mathcal{G}}(\omega, \zeta, \mathbf{k}_F, \mathbf{r}) = -i \begin{pmatrix} \vec{a} \vec{T} & \hat{0} \\ \hat{0} & \vec{a} \vec{T} \end{pmatrix}; \quad (4.6)$$

where the f, f^+ , defined by (3.1), obey equations similar to (3.2) and (3.3) but with the scattering part now reading

$$\int_{S_F} d^2 q_F \rho(\mathbf{q}_F) \{ g(\omega, \mathbf{k}_F, \mathbf{r}) (\mathcal{W}(\mathbf{k}_F, \mathbf{q}_F) - \mathcal{W}_p(\mathbf{k}_F, \mathbf{q}_F)) f(\omega, \mathbf{q}_F, \mathbf{r}) \\ - f(\omega, \mathbf{k}_F, \mathbf{r}) (\mathcal{W}(\mathbf{k}_F, \mathbf{q}_F) + \mathcal{W}_p(\mathbf{k}_F, \mathbf{q}_F)) g(\omega, \mathbf{q}_F, \mathbf{r}) \}. \quad (4.7)$$

Similarly, the functional (3.10) is changed; the factors $\mathcal{W}$ beside f and g are replaced now by $\mathcal{W} - \mathcal{W}_p$ and $\mathcal{W} + \mathcal{W}_p$ respectively. The well known effects² of paramagnetic scattering in superconductors are easily extracted from these final formulas.

* It is not claimed, of course, that this random distribution is realistic in high magnetic fields; it is considered here as a model only.

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